

A Width-Chain Estimate of the MOND Acceleration Scale from MIGHTEE-HI COSMOS, with its Flux-Scale, Line-Width and Kernel Systematics

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Abstract

The programme ties MOND’s acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived; its footings are $9.3603 \times 10^{-11} (\rho_\Lambda)$ and $1.1312 \times 10^{-10} \text{ m s}^{-2} (\rho_{\text{crit}})$. We apply a width chain, frozen before any width was read, to the MeerKAT MIGHTEE-HI COSMOS catalogue: V comes from the rest-frame W_{50} and inclination, the baryons are $1.33 M_{\text{HI}} + M_\star$ at half the H I diameter, and a median-residual estimator with the exponential kernel returns a_0 . Forty-seven golden discs at $0.027 \leq z \leq 0.093$ survive, all at $g_{\text{bar}}/a_0 \leq 0.084$, 37 gas-dominated. The catalogue paper’s worked examples show its widths are rest-frame. On the catalogue’s flux scale and $H_0 = 70$ distances $a_0 = 1.31 \times 10^{-10} \text{ m s}^{-2}$ (statistical ± 0.034 or ± 0.048 dex, by bootstrap or per-galaxy spread; recipe ± 0.128 dex), $\kappa = 0.70$ or 0.58 : consistent with SPARC and the alternative footing, with the canonical footing 0.019 dex outside the recipe width. On the $H_0 = 67.4$ distances of ρ_Λ ($a_0 = 1.21 \times 10^{-10}$) it is inside (0.111 dex away). Arecibo fluxes exceed the catalogue’s for brighter, nearer galaxies; applied to our discs’ gas they give $0.90\text{--}1.13 \times 10^{-10}$ (0.90 for the frozen pairs, 1.07 for a post hoc extrapolation), inside the recipe width of both footings. Line-width corrections lower a_0 by up to 0.10 dex; kernels without the exponential’s next-order term raise it by $0.06\text{--}0.07$ dex; dropping the S/N cut, by 0.08 dex. A SPARC closure tests only the baryon recipe. The result is not a footing discriminator and, with a lever of 0.009 dex, not an $a_0(z)$ test. The cold mass ($\Omega_c h^2 \approx 0.12$) is still required.

1 Introduction

The relation and its scale. Milgrom’s modified dynamics [M83] replaces the Newtonian acceleration below a scale a_0 . In rotating galaxies the observed centripetal acceleration g_{obs} follows the baryonic one g_{bar} along a single radial acceleration relation: 2693 points in 153 SPARC galaxies [LMS16, MLS16, L17], with SPARC’s fitted scale $g_{\dagger} = 1.20 \times 10^{-10} \text{ m s}^{-2}$ and a systematic uncertainty of ± 0.24 in the same units, 0.08 dex [MLS16]. A full joint inference over the SPARC distances, inclinations, luminosities and mass-to-light ratios finds $a_0 = (1.19 \pm 0.04 \text{ (stat)} \pm 0.09 \text{ (sys)}) \times 10^{-10} \text{ m s}^{-2}$ [D23]. From the baryonic Tully–Fisher relation (BTFR) of gas-rich galaxies, McGaugh [McG12] finds the equivalent of $a_0 = 1.3 \pm 0.3$ in the same units.

MeerKAT and line widths. The MIGHTEE-HI survey [MM26] measures H I out to $z \approx 0.09$ with one instrument and one pipeline. Ponomareva et al. [P21] built the H I BTFR of 67 early-science galaxies over $0 \leq z \leq 0.081$ from the global-profile width W_{50} (slope 3.66) and from resolved rotation curves, and found no evolution. Rajohnson et al. [R22] measured the H I size–mass relation of 204 galaxies (slope 0.501, intercept -3.252). Văršeteanu et al. measured the radial acceleration relation with resolved, SED-based mass-to-light ratios, first in 19 H I-selected COSMOS galaxies to $z = 0.08$ [V25], with $a_0 = (1.69 \pm 0.13) \times 10^{-10} \text{ m s}^{-2}$ (as restated in V26) and tentative evidence for redshift evolution, then in 130 resolved MIGHTEE-HI and LADUMA galaxies [V26]. V26 fit the *same* exponential function that we use and find $a_0 = (1.50 \pm 0.05) \times 10^{-10}$ (direct photometry) and $(1.48 \pm 0.04) \times 10^{-10}$ (Sérsic photometry); with the δ -family of interpolating functions they find $(1.86 \pm 0.06) \times 10^{-10}$. Their MIGHTEE and LADUMA galaxies alone give $a_1 = (-1.60 \pm 2.33) \times 10^{-10} \text{ m s}^{-2}$ per unit redshift; anchored at $z \approx 0$ by SPARC the same fit gives a formal 5.0σ rise of a_0 with z for mass-to-light ratios that vary between galaxies, which they attribute to the different selection of the anchor sample, and $a_1 = (-4.80 \pm 0.76) \times 10^{-10}$ per unit redshift, a formal 6.3σ fall, with a constant ratio of 0.6: the sign of the anchored trend is set by the mass-to-light treatment. For line widths, Lelli et al. [L19] give statistical relations between W_{50} and the flat rotation speed V_{flat} for SPARC, and Papastergis et al. [PAH16] measured a W_{50} -based BTFR slope of 3.75 ± 0.11 for heavily gas-dominated, edge-on ALFALFA galaxies, with a dependence on profile shape. The standard width corrections are for turbulence [VS01] and instrumental broadening [S05]. The MIGHTEE-HI COSMOS catalogue [MM26] lists 293 sources at $0.004 < z < 0.093$ with H I masses, W_{50} widths, inclinations and stellar masses.

The framework. The programme ties the scale to the vacuum energy density, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived. With $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_\Lambda = 0.6847$ [PL20] this gives the canonical footing $a_0 = 9.3603 \times 10^{-11} \text{ m s}^{-2}$; the same coefficient on the critical density gives the alternative footing 1.1312×10^{-10} . Its galaxy law is a MOND kernel ν with $g_{\text{obs}} = \nu(g_{\text{bar}}/a_0) g_{\text{bar}}$. The framework’s own closed form, $\nu = \sqrt{1 + a_0/g_{\text{bar}}}$, is the relation contained in Milgrom’s vacuum-effect paper [M99] (with a different coefficient). The measurement

lanes use the monotone kernel ν_{mono} , which below $g_{\text{bar}}/a_0 = 2.54$ equals the exponential form $\nu = [1 - \exp(-\sqrt{y})]^{-1}$ ($y = g_{\text{bar}}/a_0$) to 2×10^{-4} ; that form is the fitting function of [MLS16] and V26 and the $\alpha = \frac{1}{2}$ member of the family of Milgrom & Sanders [MS08]. The footings are numbers fixed by κ and the density and carry no kernel, but any a_0 inferred from data does: at the accelerations of this sample the inferred value depends on the kernel (Section 5), so every comparison with a footing below is conditional on the exponential kernel, and we give the closed form alongside. The distinctive prediction is a constant $a_0(z)$; the rival reading $a_0 \propto H(z)$ rises by a factor $E(z)$. The framework adds no dark-matter particle, but the cold mass ($\Omega_c h^2 \approx 0.12$) is still required for cosmology.

This paper. We report a catalogue-based estimate of the local a_0 from MIGHTEE-HI (lane CFG301), with criteria frozen and committed before any width, velocity or mass value was read, a re-measurement of the inputs from the released cubes (CFG302), a check of the HI flux scale against Arecibo single-dish fluxes (CFG304) and a test of the velocity frame of the catalogue widths (CFG309). No halo model enters; the only cosmological input is the catalogue’s luminosity distance. Each lane is a folder of the public repository (Data availability). *Revision history.* This is the first deposited version. Internal revision 1.0 (not deposited) read the catalogue widths as observed-frame and divided them by $1 + z$. A referee (CFG306) showed from the catalogue paper that they are rest-frame, and a frozen follow-up lane (CFG309) confirmed it. Revision 1.1 made the rest-frame reading primary, which raises the pooled a_0 by 0.098 dex; it also replaced the flux-scale range, added the kernel, width-side and selection systematics, and corrected the description of prior work. Revision 1.2 answers a second referee (CFG310): the single-dish range now includes the frozen reading, every footing statement is given on both distance conventions and with two estimates of the statistical error, and the selection term enters the abstract. The revision 1.0 numbers (“version 1.0 frame” below) are kept as a disclosed variant.

2 Data

The catalogue. The MIGHTEE-HI COSMOS catalogue [MM26] gives, per source: z_{HI} ; the luminosity distance D_L (flat Λ CDM, $H_0 = 70$, $\Omega_M = 0.3$, no peculiar-velocity correction); $\log M_{\text{HI}}$; the 3-D signal-to-noise SN_{3D} ; W_{50} from a busy-function fit [W14] at 50% of the average of the two peaks, not corrected for inclination; the inclination from g -band isophotes with intrinsic thickness $q_0 = 0.2$; the stellar mass from BAGPIPES SED fits [C18]; and five quality flags (low confidence, blended, confused, bad ellipse, contaminated). The *golden sample* has all five flags zero. The catalogue’s distances agree with flat Λ CDM at $H_0 = 70$ to 0.45% over the survivors, and its inclinations follow from its axis ratios with $q_0 = 0.2$ to 0.8° (rounding; this paper’s script).

The frozen cut. Applied by code, in this order, before any width was read as data: the golden sample (293 to 202 sources); $0.02 \leq z_{\text{HI}} \leq 0.093$ (188; a 300 km s^{-1} peculiar velocity is then at most 5% of the distance); $45^\circ \leq i \leq 80^\circ$ (122); $\log M_{\text{HI}} \geq 9.0$ (78); $W_{50} \geq 80 \text{ km s}^{-1}$ with $\sigma_{W_{50}}/W_{50} \leq 0.15$ (70); $SN_{3D} \geq 8$ (47); finite inputs (47). The 47 survivors are split by redshift into three equal-count windows: W1 (16 galaxies, z 0.0266–0.0476, median 0.042, median D_L 185 Mpc), W2 (16, z 0.0506–0.0756, median 0.067, 300 Mpc) and W3 (15, z 0.0773–0.0930, median 0.081, 368 Mpc). The median gas fraction is 0.70.

3 Method

The chain. For each galaxy,

$$V = \frac{W_{50} - \delta}{2 \sin i}, \quad M_b = 1.33 M_{\text{HI}} + M_\star, \quad R = \frac{1}{2} D_{\text{HI}}, \quad \log_{10} \frac{D_{\text{HI}}}{\text{kpc}} = 0.506 \log_{10} \frac{M_{\text{HI}}}{M_\odot} - 3.293,$$

$$g_{\text{bar}} = \frac{GM_b}{R^2}, \quad g_{\text{obs}} = \frac{V^2}{R}, \quad D = \frac{g_{\text{obs}}}{g_{\text{bar}}},$$

with W_{50} the catalogue’s rest-frame width (Section 5), $\delta = 0$ (the primary; a turbulence term $\delta = 11 \text{ km s}^{-1}$ is a recipe knob), the catalogue inclination, the factor 1.33 for helium, no molecular gas, and the BAGPIPES $M_\star$. CFG301 froze the chain with a division of W_{50} by $1 + z$; CFG309 re-ran CFG301’s committed script with that one factor removed (Section 5). The size relation has the form of Wang et al. [W16]; its coefficients are those adopted in the programme’s earlier width lane (CFG260) and were not re-checked against the text of [W16], whose abstract gives only the scatter (0.06 dex). All baryons are placed inside R as a point mass, so g_{bar} uses the total M_b : an asymptotic approximation, not the mass enclosed at R . The radius that W_{50} samples is HI-weighted and usually smaller than R_{HI} , so the D_{HI} knob below may understate the radius uncertainty.

The estimator. The implied ratio s^* is the root of $\text{median}_i \log_{10}[D_i/\nu(g_{\text{bar},i}/(a_0 s))] = 0$ with $\nu = \nu_{\text{mono}}$ and a_0 the canonical value; the measured scale is $s^* \times 9.3603 \times 10^{-11} \text{ m s}^{-2}$ (the alternative footing gives the same absolute a_0 to $3 \times 10^{-17} \text{ dex}$). Intervals come from a bootstrap over galaxies ($B = 4,000$). In the deep limit $\nu \rightarrow y^{-1/2}$ and the estimator reduces to $a_0 = V^4/(GM_b)$, independent of R ; at the next order a galaxy on the law shows $V^4/(GM_b) = a_0 y \nu(y)^2 > a_0$, and the estimator removes that factor. The exponential kernel behaves as $\nu \approx y^{-1/2} + \frac{1}{2}$ at small y ; kernels without the $+\frac{1}{2}$ return a larger a_0 from the same data.

The recipe half-width is the quadrature of four knobs, each applied to every galaxy: $\delta = 11 \text{ km s}^{-1}$; $M_* \pm 0.25 \text{ dex}$; D_{HI} (hence R) $\pm 0.15 \text{ dex}$; and the distances moved from $H_0 = 70$ to 67.4 (the value behind ρ_Λ). The isotropic $\sin 60^\circ$ in place of the catalogue inclinations is reported as a sensitivity. A second statistic, frozen with the first, is the direct ratio $a_0^{\text{BTFR}} = V^4/(GM_b)$ per galaxy, with its median over (i) all survivors, (ii) the gas-dominated discs ($1.33 M_{\text{HI}} > M_*$) and (iii) the deep discs ($y < 0.5$).

Calibration controls (frozen; each could fail). The values below are at the rest-frame reading (CFG309); all four also pass in the version 1.0 frame.

- *CC1, local level:* the nearest window W1 must return s^* within $\pm 0.15 \text{ dex}$ of the SPARC scale ($1.20/0.93603 = 1.282$) and of the programme’s ALFALFA control (CFG281, $s^* = 1.449$ [H18, D20]), with a recipe half-width $\leq 0.20 \text{ dex}$. W1 gives $s^* = 1.440$: $+0.050 \text{ dex}$ from 1.282, -0.003 dex from 1.449, half-width 0.135. PASS. CFG281 is itself *not calibrated* by its own record and uses $\delta = 9 \text{ km s}^{-1}$, observed-frame ALFALFA widths and the isotropic $\sin 60^\circ$, so it is not the same recipe; the $\pm 0.15 \text{ dex}$ tolerance is also wider than the 0.082 dex between the footings.
- *CC2, SPARC closure of the baryon side:* the same M_b (with $0.5 L_{[3.6]}$ for the stars), size relation and point mass applied to 123 SPARC galaxies (quality ≤ 2 , $i \geq 30^\circ$, $V_{\text{flat}} > 0$) with $g_{\text{obs}} = V_{\text{flat}}^2/R$ return $s^* = 1.304$, $+0.008 \text{ dex}$ from 1.282. With the measured R_{HI} in place of the relation the value is 1.324; the relation is unbiased on SPARC (median -0.002 dex). PASS. CC2 uses V_{flat} and SPARC’s own M_{HI} ; it tests neither W_{50} , the frame, δ nor the MeerKAT flux scale.
- *CC3, drift:* $|\log s^*(W3) - \log s^*(W1)| \leq 0.10 \text{ dex}$. The drift is $+0.022 \text{ dex}$. PASS, with little power: the statistical width of the drift is 0.141 dex.
- *CC4, self-test:* in 100 worlds with widths fabricated from the law at $s = 2$ on the survivors’ own masses and inclinations (0.15 dex scatter, 10% width errors), the bias is -0.013 dex (tolerance 0.023) and the 95% interval covers the truth in 93 of 100. PASS (run once; it tests the estimator, which the frame does not enter).

The width side, tested on SPARC (post hoc, CFG306). On 22 SPARC galaxies with an ALFALFA width (quality ≤ 2 , $i \geq 30^\circ$, ALFALFA code 1), $W_{50}/(2 \sin i)$ exceeds V_{flat} by $+0.026 \text{ dex}$ (68% $+0.022$ to $+0.032$), which in the deep limit is $+0.10 \text{ dex}$ in a_0 . The CC2 recipe run on the same galaxies with $W_{50}/(2 \sin i)$ in place of V_{flat} returns s^* higher by $+0.184 \text{ dex}$ at $\delta = 0$ and by $+0.071 \text{ dex}$ at $\delta = 11 \text{ km s}^{-1}$. The two numbers are not in conflict: the $+0.10 \text{ dex}$ is the deep-limit translation (4×0.026) that applies to our discs, all at $y \leq 0.084$, while the SPARC galaxies are not deep, and away from the deep limit the estimator amplifies a velocity offset well beyond the factor of four. The chain does not correct for this; on these numbers a correction would *lower* a_0 , and $\delta = 5 \text{ km s}^{-1}$ lowers our pooled value by 0.044 dex (Table 2). No empirical $W_{50} \rightarrow V_{\text{flat}}$ relation (such as those of [L19]) is applied as a two-sided term: [L19]’s relations are calibrated on SPARC galaxies that are more massive and less gas-dominated than ours, the 22-galaxy overlap above measures only a mean offset, and either would be a post hoc choice of a correction whose sign is already known. The width side therefore remains uncalibrated for these discs.

Reactivity and reproduction. In the version 1.0 frame, widths multiplied by 1.1892 raise the pooled s^* by $+0.322 \text{ dex}$ (expected 0.30 ± 0.05 ; MUTATE=1); $\sin i_{\text{eff}} = 0.70$ against the $\sin 60^\circ$ variant raises it by $+0.392 \text{ dex}$ (expected 0.37 ± 0.06 ; MUTATE=2). The chain, fed CFG281’s ALFALFA arrays, reproduces CFG281’s committed s^* exactly (1.332408 unmatched, 1.449173 matched; M5). All six frozen hand estimates hit. CFG309’s re-run reproduces CFG301’s committed values exactly when the factor is kept, and its rest-frame value agrees with CFG306’s independent code to $2 \times 10^{-16} \text{ dex}$.

Disclosures. Before the freeze only the catalogue’s header line had been read. The frozen text asks for mass-matched windows in CC1 and CC3 but does not define the matching; the rule fixed in the script before any value was read (W1 and W2 matched to W3’s $\log M_{\text{HI}}$ quartile bins, at least three galaxies per bin) could not be met (W1 had 8/3/2/2 galaxies per bin), so CC1, CC3 and the drift use unmatched windows. W1’s median $\log M_{\text{HI}}$ (9.605) is 0.155 dex below W3’s (9.76). A reader who insists on matching should treat CC1 and CC3 as not evaluable. The BTFR kernel-factor diagnostic was added after the SPARC control and before the measurement, using baryons only. The rest-frame reading was adopted after the measurement, on CFG309’s frozen rule.

set	N	z median	D_L [Mpc]	s^*	a_0 [$10^{-10} \text{ m s}^{-2}$]	68%	95%	recipe [dex]
W1	16	0.042	185	1.440	1.348	1.254–1.597	1.093–2.317	0.135
W2	16	0.067	300	1.394	1.305	1.233–1.506	1.038–1.696	0.136
W3	15	0.081	368	1.513	1.416	1.130–1.965	0.655–1.967	0.132
pooled	47	—	—	1.401	1.311	1.273–1.418	1.206–1.669	0.128

Table 1: The implied a_0 per redshift window and pooled, with the catalogue widths read as rest-frame (CFG309’s re-run of CFG301’s committed chain; $s^* = a_0/9.3603 \times 10^{-11}$). The 68% and 95% columns are bootstrap intervals over galaxies, in $10^{-10} \text{ m s}^{-2}$; the recipe column is the half-width of the four-knob quadrature.

4 Results

The level. On the catalogue’s flux scale the pooled scale is $a_0 = 1.311 \times 10^{-10} \text{ m s}^{-2}$ ($s^* = 1.401$), with a bootstrap 68% interval of 1.273–1.418, a 95% interval of 1.206–1.669 and a bootstrap standard deviation of 0.034 dex (the 68% interval of a median of 47 is lumpy), and a recipe half-width of 0.128 dex (Table 1, Fig. 1). The statistical error depends on the method: the per-galaxy implied values have a robust spread of 0.261 dex, which gives an asymptotic standard error of the median of 0.048 dex (post hoc), wider than the bootstrap because the per-galaxy values cluster at and just below the median. We quote both, and every statement below holds under the larger. The scale lies +0.146 dex above the canonical footing, +0.064 dex above the alternative and +0.038 dex above SPARC’s 1.20, inside SPARC’s own 0.08 dex systematic. The canonical footing lies outside the 95% statistical interval under either error (bootstrap 1.206–1.669; $\pm 1.96 \times 0.048$ dex, 1.057–1.626); the alternative footing lies outside the bootstrap interval but inside the wider one. With the recipe half-width the alternative footing is inside and the canonical footing is outside by 0.019 dex. All of this is on the catalogue’s $H_0 = 70$ distances; on the $H_0 = 67.4$ distances of ρ_Λ the canonical footing is inside the recipe width and the alternative footing inside both statistical intervals (Section 5). The recipe knobs contribute 0.098 (δ), 0.069 (M_*), 0.026 (D_{HI}) and 0.036 dex (H_0); the $\sin 60^\circ$ sensitivity is 0.035 dex (0.132 with it in the quadrature). All 47 galaxies are deep: $y = g_{\text{bar}}/a_0$ at R has quartiles 0.028, 0.033 and 0.042 and a maximum of 0.084. In the version 1.0 frame the pooled value is 1.046×10^{-10} (−0.098 dex).

The windows and the drift. The window medians are 1.348, 1.305 and 1.416 (Table 1); their bootstrap intervals are lumpy, because a median of 15–16 galaxies jumps between galaxies. The slope of $\log_{10} s^*$ on $\log_{10} D_L$ is +0.05 per dex (68% −0.44 to +0.35; 95% −1.03 to +0.64): not resolved. The ratio W3/W1 is 1.051 ($\log_{10}$: +0.022 $\pm$ 0.141). Version 1.0’s remark that its drift had the sign of the programme’s ALFALFA control is withdrawn: the drift changed sign. Between W1 and W3 the rival $a_0 \propto H(z)$ predicts +0.009 dex, far below the statistical width. A joint fit of the per-galaxy residual in z and $\log M_{\text{HI}}$ (post hoc, this paper) gives −2.6 per unit z (68% −5.2 to −0.5; 95% −8.0 to +1.5) and +0.09 per dex of M_{HI} (95% −0.23 to +0.49): neither is resolved at 95%, although the 68% interval of the redshift term excludes zero.

The direct BTFR. The median $V^4/(GM_b)$ is $1.530 \times 10^{-10} \text{ m s}^{-2}$ over all 47 (68% 1.460–1.725; +0.213 dex from the canonical footing, +0.131 dex from the alternative; recipe 0.111 dex) and 1.464×10^{-10} over the 37 gas-dominated discs (68% 1.457–1.530; +0.194 and +0.112 dex; recipe 0.120 dex). The deep subset (iii) is the same 47 galaxies as (i). These rows sit above the s^* row by design: computed from the baryons alone, the median kernel factor $y\nu_{\text{mono}}(y)^2$ is 1.197 (+0.078 dex) at the canonical a_0 , 1.172 at 1.20×10^{-10} and 1.183 for the gas-dominated discs. The gas-dominated median differs from the full one by −0.019 dex, but the per-galaxy residual correlates with the gas fraction and with $\log M_*$ (Section 5), so the stellar mass is not negligible. The inverse BTFR slope of the 47 is 3.43 (68% 3.11–3.79; post hoc), against 3.66 [P21] and a forward-equivalent 3.73 [V26]; with a slope below 4 the median $V^4/(GM_b)$ depends on the sample’s mass distribution. The same recipe on SPARC gives a BTFR level of 1.526×10^{-10} (CC2’s diagnostic, with V_{flat}). Fig. 2 shows the 47 discs against the deep-MOND line on both footings.

The coefficient. Read as $\kappa = \frac{1}{2} a_0/a_0^{\text{footing}}$ (with $\kappa = \frac{1}{2}$ itself fitted), the pooled scale gives $\kappa = 0.700$ on the ρ_Λ footing (bootstrap 68% 0.680–0.758, 95% 0.644–0.892; scaled by $10^{\pm 0.128}$, 0.52–0.94) and $\kappa = 0.580$ on ρ_{crit} (68% 0.563–0.627, 95% 0.533–0.738; recipe 0.43–0.78). On statistics alone $\kappa = \frac{1}{2}$ lies below the 95% interval on the ρ_Λ footing under either error (with the per-galaxy error, 0.565–0.868); on ρ_{crit} it lies below the bootstrap interval but inside the per-galaxy one (0.467–0.719). With the recipe width it is just outside on ρ_Λ and inside on ρ_{crit} . These are on the catalogue’s $H_0 = 70$ distances. On the $H_0 = 67.4$ distances that match ρ_Λ , the scale is 1.208×10^{-10} and the central values become 0.645 and 0.534 (Section 5); $\kappa = \frac{1}{2}$ is then inside the recipe width on both footings and inside both statistical intervals on ρ_{crit} , and on ρ_Λ it stays below both (bootstrap 0.594–0.823; per-galaxy 0.520–0.800).

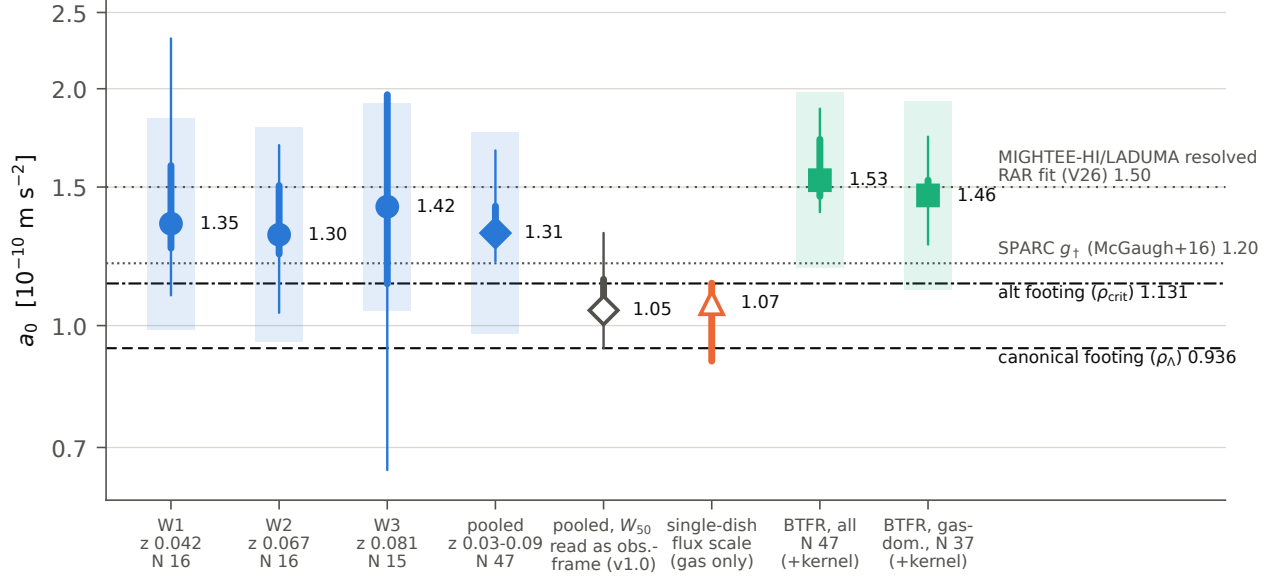


Figure 1: The implied a_0 per redshift window (circles) and pooled (filled diamond), with the catalogue widths read as rest-frame, bootstrap 68% (thick) and 95% (thin) intervals and the recipe half-width (shaded box). Open grey diamond: the pooled value in the version 1.0 frame (1.05). Open triangle: the pooled value with the gas on the Arecibo single-dish flux scale at the post hoc extrapolation in SN_{3D} (1.07); its bar, 0.90–1.13, runs from CFG304’s frozen code-1 reading to the extrapolation in z (Section 5). Squares: the direct BTFR medians, which include the kernel’s next-order factor (+0.078 dex). Lines: the canonical (0.936) and alternative (1.131) footings, SPARC’s fitted g_t (1.20) and the MIGHTEE-HI/LADUMA resolved fit (1.50), in $10^{-10} \text{ m s}^{-2}$.

Against other determinations. The pooled scale is +0.042 dex from the joint inference of [D23] (1.19) and -0.058 dex from the MIGHTEE-HI/LADUMA resolved fit of [V26] (1.50), which uses the same exponential function with resolved rotation curves and resolved SED mass-to-light ratios; their sub-samples span 1.31–1.80 (CFG279), and their 19-galaxy COSMOS fit gave 1.69 [V25]. The two MIGHTEE analyses are not the same measurement: ours places all baryons at $D_{\text{HI}}/2$ and reads the velocity from W_{50} , and at most 28 of V26’s 130 galaxies are in COSMOS (V25’s 19 and 9 from the new catalogue). V26’s HI comes from 3D Barolo fits to the MeerKAT cubes with a smoothed 3σ source mask, so both analyses carry MeerKAT’s interferometric flux scale.

5 Systematics

The velocity frame (CFG309). Under criteria frozen before the catalogue paper’s text was read for this purpose, CFG309 found the following. (T1) The catalogue paper never states the frame in words. (T2) Its five worked example spectra print each W_{50} both in km s^{-1} and in 26.126 kHz channels; in all five the km s^{-1} per channel equals $c \Delta\nu/\nu_{\text{obs}}$, the rest-frame width, to +0.066 to +0.068% (spread 1.4×10^{-5}), and the observed-frame and radio conventions are excluded in all five. Four of the five printed widths equal the catalogue’s values; the fifth (J100404.9+014303: $237.858 \text{ km s}^{-1}$ printed, 223 catalogued) is an unexplained anomaly. (T3) The ALFALFA comparison cannot separate the frames (a lever of 0.012 dex). (T4) Busy-function fits to the released cubes lean towards the observed frame (median offset -0.017 dex for the rest-frame reading), but the test failed its injection control (rest-frame worlds recovered as rest-frame in 13 of 20) and is inadmissible. The decision therefore rests on the catalogue paper’s own worked numbers; no independent data test could decide it. CFG302’s raw widths (rest-frame) agree with the catalogue’s to -0.022 dex and would agree to +0.002 dex under the observed-frame reading, which does not decide it either. All three data tests lean the same way, weakly, towards the observed frame: CFG302’s raw widths (+0.002 against -0.022 dex), the ALFALFA width ratio of CFG304 (zero, the observed-frame prediction, against -0.012 dex for the rest frame; Section 5) and the inadmissible busy-function fits. None decides, but together they keep the observed-frame variant (1.046×10^{-10} ; Table 2, Fig. 1) a live alternative. Version 1.0’s $(1+z)$ division came from the programme’s ALFALFA lanes, whose widths are observed-frame.

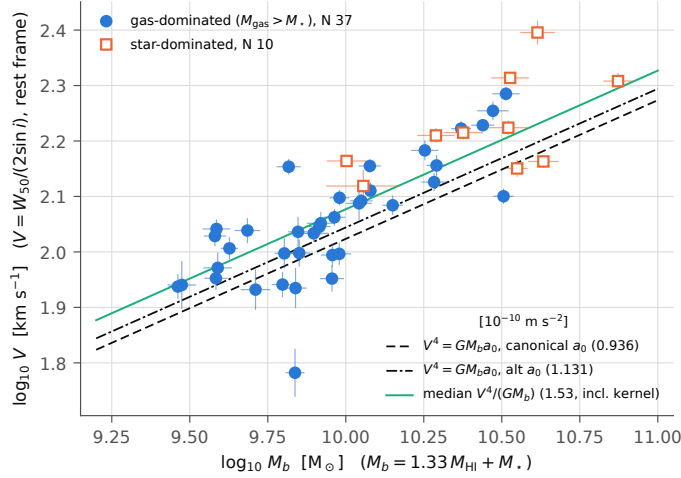


Figure 2: The 47 discs in the plane of $V = W_{50}/(2 \sin i)$ (rest-frame widths) and $M_b = 1.33 M_{\text{HI}} + M_*$: gas-dominated (filled, 37) and star-dominated (open, 10). Error bars: the catalogue’s W_{50} error and the quadrature of its M_{HI} and M_* errors; inclination errors are not tabulated and not shown. Lines: $V^4 = GM_b a_0$ on the canonical and alternative footings (the deep-MOND limit) and the median $V^4/(GM_b) = 1.53 \times 10^{-10} \text{ m s}^{-2}$, which includes the kernel factor.

variant (all on the same 47 discs unless stated)	$\Delta \log_{10} a_0$ [dex]	source
H I flux scale, single dish, gas alone (R follows): ratio extrapolated in SN_{3D} to our discs (-0.115 dex)	-0.089	CFG306 P3 / post hoc
same, the 7 matched pairs that pass CFG301’s cut (-0.159) / ratio extrapolated in z (-0.087)	$-0.132 / -0.064$	CFG306 P3 / post hoc
same, the 15 code-1 pairs unadjusted (-0.195)	-0.163	CFG306 P3 / post hoc
velocity frame: W_{50} divided by $1+z$ (version 1.0)	-0.098	CFG301
line width: $\delta = 5 / 11 \text{ km s}^{-1}$	$-0.044 / -0.098$	post hoc / CFG309
kernel: simple / standard ($n = 2$)	$-0.001 / +0.063$	post hoc (= CFG306 P2)
kernel: closed form $\sqrt{1 + 1/y} / \delta$ -family ($\delta = 4.1$)	$+0.058 / +0.067$	post hoc (= CFG306 P2)
selection: $SN_{3D} \geq 6 / \geq 10 / \geq 12$ in place of 8	$+0.034 / +0.006 / +0.034$	post hoc
selection: no SN_{3D} cut (70 discs)	$+0.084$	post hoc
molecular gas $M_{\text{H}_2} = 0.1 / 0.3 M_{\text{HI}}$	$-0.025 / -0.073$	post hoc (= CFG306 P7)
inclination: $q_0 = 0.10 / 0.30$ in place of 0.2	$+0.029 / -0.050$	post hoc
inclination: isotropic $\sin 60^\circ$	$+0.035$	CFG309
stellar mass ± 0.25 dex	$-0.076 / +0.063$	CFG309 recipe
$D_{\text{HI}} \pm 0.15$ dex	$+0.021 / -0.030$	CFG309 recipe
MIGHTEE’s own size relation [R22]	-0.001	post hoc
distances on $H_0 = 67.4$ instead of 70	-0.036	CFG309 recipe

Table 2: Systematic variants of the pooled a_0 (1.311×10^{-10} , rest-frame widths). Post hoc rows are this paper’s script (`make_paper40_figures.py`), which first reproduces CFG301’s and CFG309’s committed values exactly and CFG306’s flux and kernel rows; they were not frozen. The flux-scale rows are a signed systematic from galaxies brighter and nearer than ours, not a measurement.

The H I flux scale (CFG302). CFG302 re-measured the 188 golden sources at $0.02 \leq z \leq 0.093$ from the four released MIGHTEE-HI r1p0 cubes (circular restoring beam $62.2\text{--}77.3''$ over the valid channels, median $75.9''$; per-sub-cube medians $74.7\text{--}77.3''$; $8''$ pixels), with criteria frozen before any spectrum was extracted. Of the 179 sources in its primary set, 58 reach its frozen detection threshold. *The widths agree:* the median log ratio of our rest-frame W_{50} to the catalogue’s is -0.022 dex with a robust scatter of 0.039 dex (4 of 58 pull outliers). This confirms the catalogue’s W_{50} *measurement*, not the conversion of W_{50} into a rotation speed. *The fluxes do not:* the median ratio over all 179 is 0.22 , and over the 58 detections the median log ratio is -0.30 dex (a ratio of 0.50 , scatter 0.24 dex; 46 of 58 pull outliers). The aperture is complete ($S(2.5\theta)/S(1.5\theta) = 1.002$), and the stacked line holds 0.60 ± 0.05 of the catalogue’s mean flux. One frozen control failed and is kept: the line-free signal-to-noise spread is 0.489 , not about 1, because the noise is anticorrelated over long spectral lags, as if the released spectra were high-pass filtered. For exactly our 47 discs (post hoc join with CFG302’s committed table), 45 are in its primary set and 20 reach its threshold; their median cube-to-catalogue flux log ratio is -0.443 dex, while their widths agree (-0.017 dex).

An independent flux scale (ALFALFA, CFG304). CFG304 matched the catalogue with the ALFALFA catalogue of Arecibo single-dish fluxes [H18], under criteria frozen before any per-galaxy flux was read: 23 matches, 15 of them ALFALFA code 1. The matches are real: shuffled positions give 0.275 matches on average. The catalogue and ALFALFA widths have a median log ratio of $+0.000$ dex, which sits on exact ties ($68\% -0.025$ to 0.000); the rest-frame reading predicts -0.012 dex and the observed-frame reading zero, so this does not decide the

frame. On the 15 code-1 pairs the median log ratio of catalogue to single-dish flux is -0.195 dex (68% -0.250 to -0.156), a ratio of about 0.64; for the 7 of them with a cube detection the cube ratio is -0.308 dex. Without pairs that have a catalogued neighbour within $3.5'$ the two ratios are -0.175 and -0.383 . The catalogue's $\log M_{\text{HI}}$ sit 0.22 dex below ALFALFA's for the same galaxies. The frozen decision is “neither”: the single dish sees more H I than both MeerKAT scales, and the cube scale lies further from it than the catalogue's, so the cube scale is not the relevant systematic. Its cause is open; the catalogue paper's iterated 3σ moment-map mask is a plausible one, not demonstrated.

How far the offset carries to our discs (post hoc, CFG306). The matched pairs are not like our discs. The 15 code-1 pairs have median z 0.028 (ours 0.065), SN_{3D} 18.7 (ours 12.4) and H I diameter $66''$ (ours $40''$); only 30% of our discs lie inside the pairs' redshift range, and 7 of the 47 have a pair. Within the code-1 pairs the ratio becomes less negative at lower SN_{3D} (Spearman -0.62 , p 0.01), and linear trends carried to our medians give -0.115 (in SN_{3D}) and -0.087 (in z , all 23 pairs); CFG304's $z \geq 0.02$ subsets give -0.137 (19 pairs) and -0.120 (confusion-clean). The direction is robust: SPARC's own M_{HI} sit $+0.053$ dex above ALFALFA's for 35 common galaxies at the same distance, so the catalogue is about 0.25 dex below the H I scale of CC2's calibrator; confusion would need about 500 times the mean cosmic density of uncatalogued H I in an Arecibo beam; and MeerKAT's 29 m shortest baseline recovers scales of about $25'$, far larger than our discs.

What the flux scale does to a_0 . The chain uses the catalogue's M_{HI} . An H I flux error changes the gas, not the BAGPIPES stellar masses, so the reading is the gas alone raised, with R following the size relation (CFG306 P3, reproduced exactly here). CFG304's frozen reading, the 15 code-1 pairs unadjusted, gives $a_0 = 0.90 \times 10^{-10} \text{ m s}^{-2}$; the 7 matched pairs that pass CFG301's own cut give 0.97, the $z \geq 0.02$ subsets 1.02 and 1.06, the ratio carried to our discs in SN_{3D} 1.07 and the z extrapolation 1.13 (all $\times 10^{-10}$). We take 0.90–1.13 as the single-dish range for our discs and give no single central value: 0.90 is the only frozen reading, and 1.07, which Fig. 1 marks, is a post hoc linear extrapolation over 15 pairs. At 0.90 κ is 0.481 (ρ_{Λ}) or 0.398 (ρ_{crit}), with the canonical footing -0.016 dex and the alternative -0.099 dex away; at 1.07 κ is 0.571 or 0.472, with the canonical footing $+0.058$ dex and the alternative -0.025 dex away. Across the range both footings are inside the recipe width. Version 1.0's reading with all baryons raised ($0.60\text{--}0.75 \times 10^{-10}$ in that frame) is withdrawn as a flux systematic: it scales the stellar masses with the H I flux. Across the two flux scales the honest range is about 0.9 to 1.31×10^{-10} , and on present evidence the catalogue-scale value is the least consistent with the single dish, not the top of a range.

Kernel. At $y \approx 0.03$ the inferred a_0 depends on the kernel. The exponential and simple forms carry the $+\frac{1}{2}$ term and agree (-0.001 dex); the standard $n = 2$ form, the framework's closed form $\sqrt{1 + 1/y}$ and V26's best δ -family member lack it and raise a_0 by $+0.06$ to $+0.07$ dex, to 1.50 with the closed form (1.22 in the version 1.0 frame). This is comparable to the 0.082 dex between the footings. Against CC2 run with the same kernel, the MIGHTEE value moves only between $+0.001$ and $+0.032$ dex, so the comparison with SPARC is robust to the kernel while the comparison with the footings is not. V26 see the same effect on MIGHTEE data (1.50 against 1.86).

Line width, inclination, stellar mass, molecular gas and size. No turbulence or instrumental correction is applied; $\delta = 11 \text{ km s}^{-1}$, the largest recipe term, lowers a_0 by 0.098 dex, and the SPARC comparison above suggests a correction of that sign. The catalogue's inclinations follow from its axis ratios with $q_0 = 0.2$; $q_0 = 0.10$ or 0.30 moves the pooled a_0 by $+0.029$ or -0.050 dex (post hoc) and the isotropic $\sin 60^\circ$ by $+0.035$ dex. The 45° cut keeps the face-on end out; optical inclinations can differ from the H I disc's, and seeing affects the axis ratios of the smallest discs. The per-galaxy residual shows no trend with inclination (Spearman -0.08) or axis ratio ($+0.09$). The stellar mass moves a_0 by $-0.076 / +0.063$ dex for ± 0.25 dex. Molecular gas at 0.1 or 0.3 of M_{HI} , omitted by the chain, lowers a_0 by 0.025 or 0.073 dex. In the deep limit the estimator does not depend on R , so the size relation enters only at next order: $D_{\text{HI}} \pm 0.15$ dex moves a_0 by 0.026 dex (half-width), and MIGHTEE's own relation [R22] by -0.001 dex (D_{HI} smaller by 0.008 dex at the median $\log M_{\text{HI}} = 9.73$). The point mass at $D_{\text{HI}}/2$ is a recipe choice, not a resolved mass model.

Selection. The frozen $SN_{3D} \geq 8$ cut removes 23 of 70 discs. Alone, those 23 give 2.23×10^{-10} , and all 70 give 1.59×10^{-10} ($+0.084$ dex; post hoc; in the version 1.0 frame 1.72 and 1.23, CFG306). Between thresholds of 6 and 12 the pooled value moves by -0.009 to $+0.034$ dex. Over the 70, the catalogue's SN_{3D} scales as $W_{50}^{-0.50}$ at fixed flux (CFG306), so the cut favours narrow lines; low-S/N fluxes or widths may also be biased. The per-galaxy residual correlates with W_{50} (Spearman $+0.63$), with the gas fraction (-0.45 , p 0.002) and with $\log M_*$ ($+0.36$, p 0.01), but not with z (-0.11) or $\log M_{\text{HI}}$ ($+0.04$). The sample is H I-flux-selected; V26 show that for such a sample the BTFR fitting direction moves their evolution term from -11.4 ± 1.3 to -8.1 ± 2.4 (both $\times 10^{-10} \text{ m s}^{-2}$ per unit z).

Distances and H_0 . The catalogue's distances assume $H_0 = 70$ and no peculiar velocities, while ρ_{Λ} uses $H_0 = 67.4$. Moving the distances to 67.4 raises every mass by $2\log_{10}(70/67.4)$ and lowers the pooled a_0 by 0.036 dex, to

1.208×10^{-10} ($s^* = 1.290$); against the footings this is the like-for-like value. On these distances the canonical footing is +0.111 dex away, inside the recipe width whether the H_0 knob is kept (0.128 dex) or removed because the shift has been applied as a correction (0.123 dex), and the alternative footing is +0.028 dex away. Evaluated on the same bootstrap resamples (post hoc), the 95% interval becomes 1.112–1.540, so the alternative footing is inside it; the canonical footing stays outside both statistical intervals (the per-galaxy one is 0.974–1.498). The footing verdicts therefore depend on the distance convention, and we give both. Peculiar velocities are random per galaxy and at most 5% in distance above $z = 0.02$.

Mass matching and sample size. The windows could not be mass-matched (Section 3), so the drift compares W1, whose median $\log M_{\text{HI}}$ is 0.155 dex lower, with W3. With 15–16 galaxies per window the window medians are lumpy, CC3 has little power and the drift slope is unresolved.

6 What the result can and cannot say

- **It can say:** an independent instrument and sample, with no halo model, give a deep-regime scale of $1.311 \times 10^{-10} \text{ m s}^{-2}$ on the catalogue’s flux scale and $H_0 = 70$ distances, 1.208×10^{-10} on $H_0 = 67.4$ distances, and $0.90\text{--}1.13 \times 10^{-10}$ with the gas on the single-dish scale. The catalogue-scale value is consistent with SPARC (+0.038 dex) and with the alternative footing (+0.064 dex, inside the recipe half-width of 0.128 dex); the canonical footing (+0.146 dex) is just outside it on the $H_0 = 70$ distances and inside it on $H_0 = 67.4$ (+0.111 dex) and on the single-dish scale.
- **It is not a footing discriminator.** The footings differ by 0.082 dex. The flux scale moves a_0 by -0.06 to -0.16 dex, the width side by up to -0.10 dex and the kernel by up to $+0.07$ dex.
- **It is not an $a_0(z)$ test.** At $z \leq 0.093$ the rival $a_0 \propto H(z)$ differs from a constant by 0.009 dex between W1 and W3, against a statistical width of 0.141 dex. The distinctive constant- $a_0(z)$ prediction needs $z \gtrsim 2$ and baryon calibrations near 0.1 dex [P38].
- **The calibration is partial.** CC2 closes the baryon side only; CC1 and CC3 are evaluable only on unmatched windows, and CC1’s second reference is itself not calibrated. The velocity side and the flux scale are not calibrated.
- **Not claimed:** no preference of the data for the programme’s reading over Λ CDM or over standard MOND; $\kappa = \frac{1}{2}$ is fitted, not derived; no completeness of the programme; the cold mass is still required and no dark-matter particle is introduced.

7 Conclusions

On 47 golden, deep-regime ($y \leq 0.084$) MIGHTEE-HI discs, 37 of them gas-dominated, a frozen width chain with the catalogue widths read as rest-frame gives $a_0 = 1.311 \times 10^{-10} \text{ m s}^{-2}$ (statistical ± 0.034 dex by bootstrap or ± 0.048 dex from the per-galaxy spread; recipe ± 0.128 dex) on the catalogue’s flux scale and $H_0 = 70$ distances; $\kappa = 0.70$ (ρ_Λ) or 0.58 (ρ_{crit}). On $H_0 = 67.4$ distances it is 1.208×10^{-10} . The frame rests on the catalogue paper’s worked examples; independent data tests could not decide it, and all three lean weakly towards the observed-frame variant. With the gas on the Arecibo single-dish scale the value lies in $0.90\text{--}1.13 \times 10^{-10}$ (0.90 the frozen reading). Width corrections would lower it by up to 0.10 dex, kernels without the exponential’s next-order term would raise it by 0.06–0.07 dex, and dropping the S/N cut would raise it by 0.084 dex. The result is consistent with SPARC and the alternative footing; the canonical footing is just outside the recipe width on the catalogue scale with $H_0 = 70$ distances, and inside it with $H_0 = 67.4$ distances or on the single-dish scale. It neither separates the footings nor tests $a_0(z)$. Total-power fluxes and a $W_{50}\text{-to-}V_{\text{flat}}$ calibration for these discs themselves would remove the two largest systematics.

Data availability

All scripts, outputs and frozen criteria are in the programme’s public repository; each lane label is a folder of `campaign_fresh_gravity/`. `CFG301_mightee_hi_catalogue_width_chain/` is the frozen chain (criteria commit 2555ab142, results 6b10c01c0); `CFG302_mightee_cube_raw_widths/` the cube re-measurement (criteria 7d317dd4f, results a89ba33b9); `CFG304_mightee_flux_scale_alfalfa/` the ALFALFA flux comparison (criteria 336b36f8f, results 34e40dac6); `CFG306_paper40_referee/` the referee report on version 1.0 and its diag-

nostics (45c41e887); CFG309_mightee_width_frame/ the width-frame test and the rest-frame re-run (criteria ec54de4ac, results 53f8fa937); CFG310_second_referee/ the second referee report (ffcd1068). Earlier lanes cited for context: CFG260_budhies_a0_z02/ (size-relation coefficients), CFG279_mightee_published_values/ (V26’s published numbers) and CFG281_budhies_local_control/ (the ALFALFA local control, built on the committed ALFALFA tables [H18, D20] in data_assembly/alfalfa_sdss_local_control/). The catalogue is the MNRAS supplementary file of [MM26], downloaded by the author in a browser because the publisher’s bot challenge blocks scripted fetches (zip sha256 prefix eb247d5de07eb30e) and committed unmodified as data_assembly/mightee_hi_catalogue_2026-10-02/MIGHTEE_HI_COSMOS_catalogue.csv (75,999 bytes, sha256 bcf9e8558bc5644805b9acb288372c17daffac76d8be6c89d7db9843af9fbf23; commit 19adffd49). The MIGHTEE-HI DR1 cubes are at the SARA DOI 10.48479/jkc0-g916 and are not in the repository. This paper’s script, post hoc numbers and audit are in qwen_claude_field_theory/papers_2026/.

Use of AI tools

Large language models from several providers assisted with derivations, data handling, code and drafting: Anthropic’s Claude (mainly via Claude Code), OpenAI models, DeepSeek, Zhipu AI’s GLM, Alibaba’s Qwen and Google’s Gemini. The author reviewed all of the content and takes responsibility for it.

Reproducibility

make_paper40_figures.py re-implements the CFG301 chain and writes nothing unless it reproduces CFG301’s committed values in the version 1.0 frame, CFG309’s committed values in the rest frame (pooled and window s^* , the H_0 , δ and M_* knobs, the BTFR medians) and CFG306’s flux and kernel rows, re-applies the frozen cut to recover the 47 discs, and reproduces the committed pooled bootstrap exactly before re-using its resamples on $H_0 = 67.4$ distances (33 checks, all exact or within 10^{-8}). Every number carried by an audit row of PAPER40_audit.py is re-read from a committed file at git HEAD or, for the post hoc rows, from PAPER40_figures_numbers.json; it is checked to be printed in the tex block carrying its tag, and plain numbers are also checked against the text of the built PDF; two mutation modes must fail. Literature values that no committed file carries are checked only for being printed. One provenance note (CFG306): CFG301’s committed stage A, SELFTEST, CC2 and M5 outputs were written by a script whose docstring later gained one line; re-running the committed script reproduces them identically.

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